

Colliding Fireworks

The sketch will consist of randomized fireworks (varying in different types, color and size). When they collide they will engage a secondary explosion, potentially causing chain reactions of fireworks to fill the screen.

In depth description:

The depictions of the fireworks will be less real and more pixel based, highlighting the emerging sparks pathways and having a clearer contrast from the background.

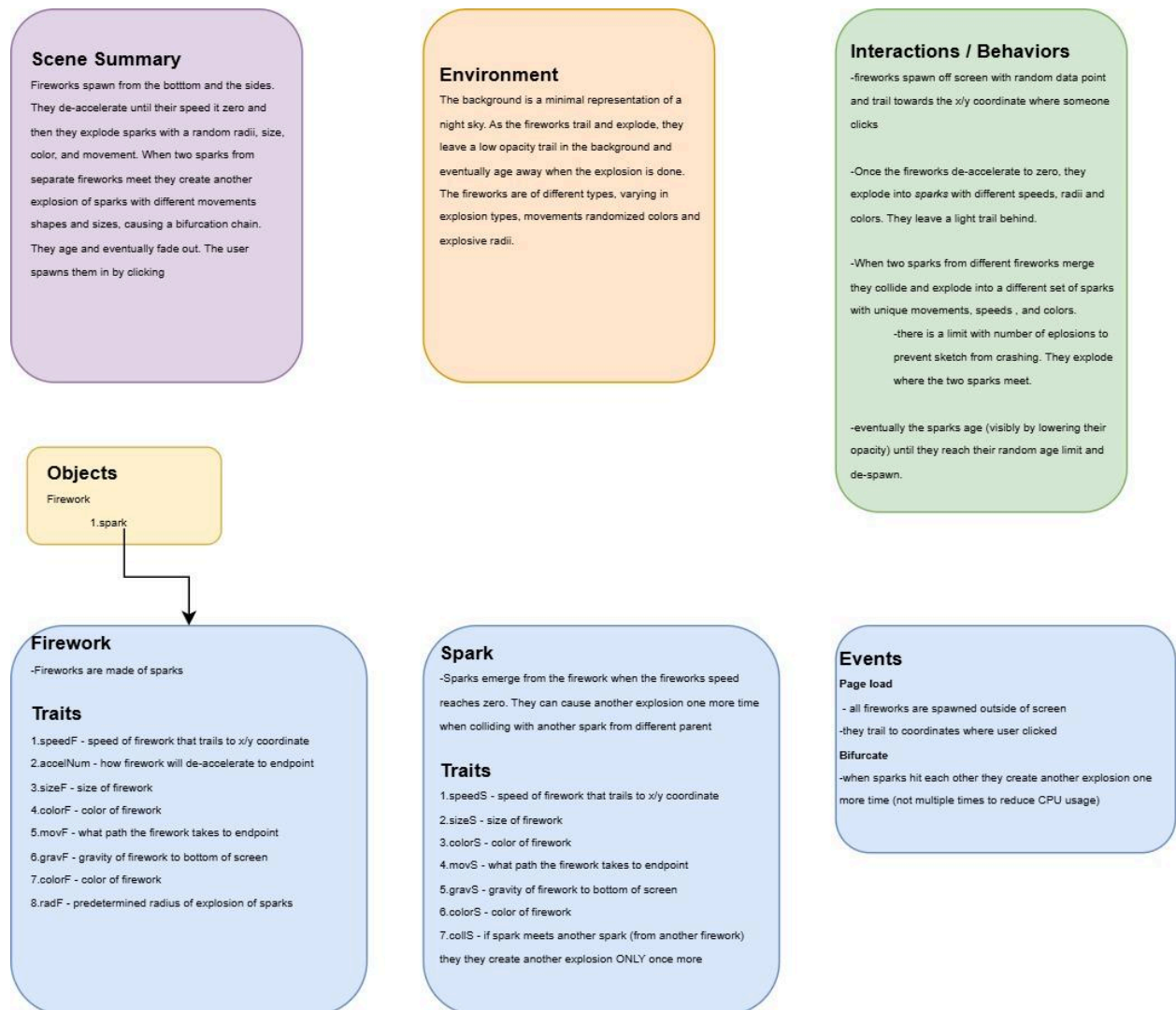
When the program starts there will only be a few fireworks launching at a time to prevent the sketch from crashing if the chain reactions cannot be halted. Instead, the user will be able to tap the screen and the fireworks will launch and explode at that x/y coordinate.

There will be a list of different classes of fireworks with different properties, ie. explosion radii, length of sparks, movement of sparks, color, varying initial tail paths, etc. Each one will also have gravity, emerging from the bottom of the screen and de-accelerating until their speed is zero, then the explosion occurs. Their radius, color, and length of each spark will be random as well as their path. The initial path of the firework will emerge from different sides as well with their own varying arcs into the scene (but will never come from max height)

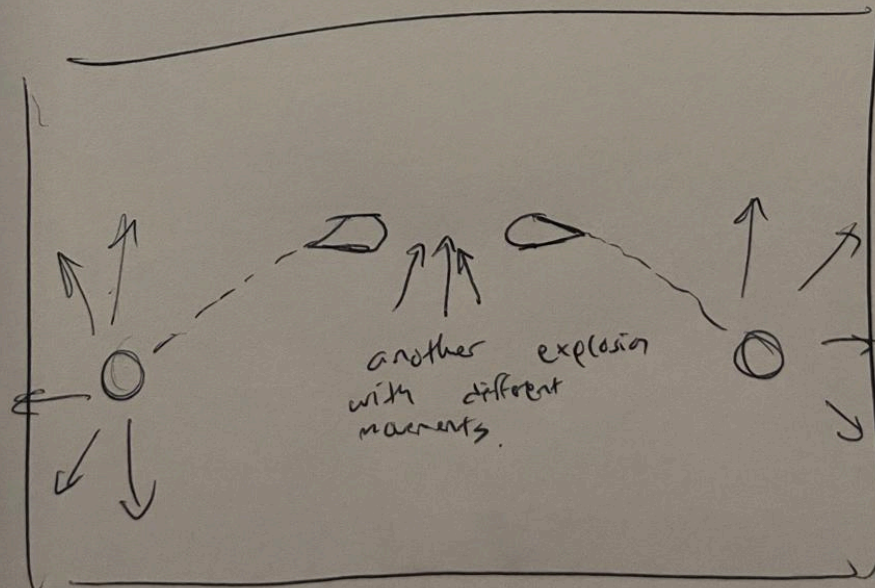
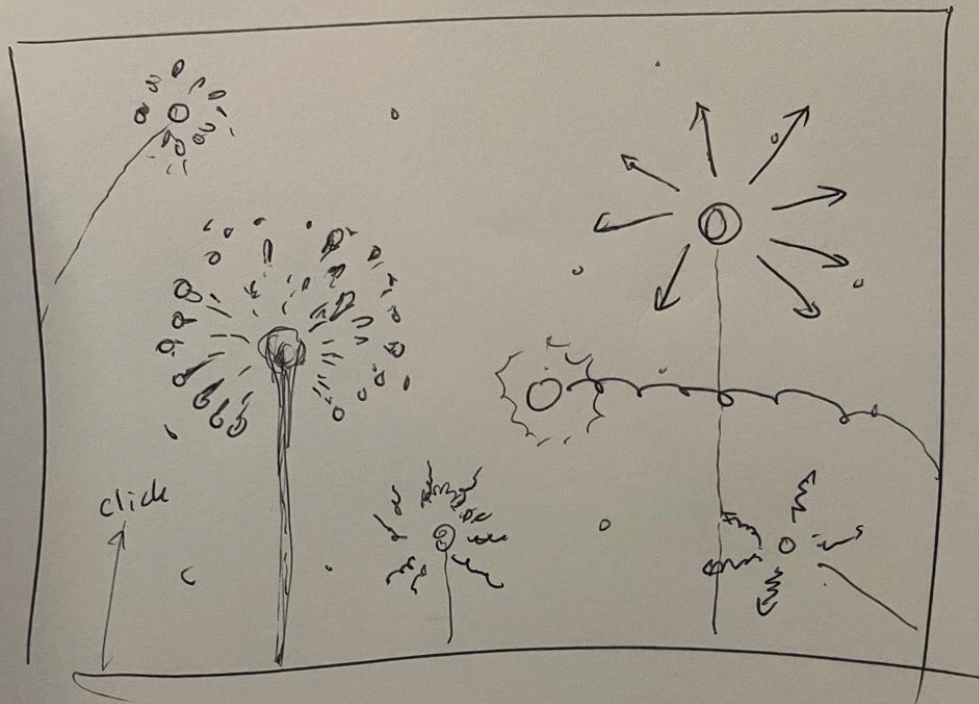
When one spark touches another spark from another firework, a secondary explosion of varying paths and radii will bifurcate at the convergence point between the two. The bifurcation points will induce explosions that potentially follow a similar random walker pathway.

When the program starts there will only be a few fireworks launching at a time to prevent the sketch from crashing if the chain reactions cannot be halted. Instead, the user will be able to tap the screen and the fireworks will launch and explode at that x/y coordinate. The fireworks and time of explosion will also have an aging property, both fading out the fireworks from the scene with opacity and also eventually die (be removed from the sketch) to conserve CPU processing power.

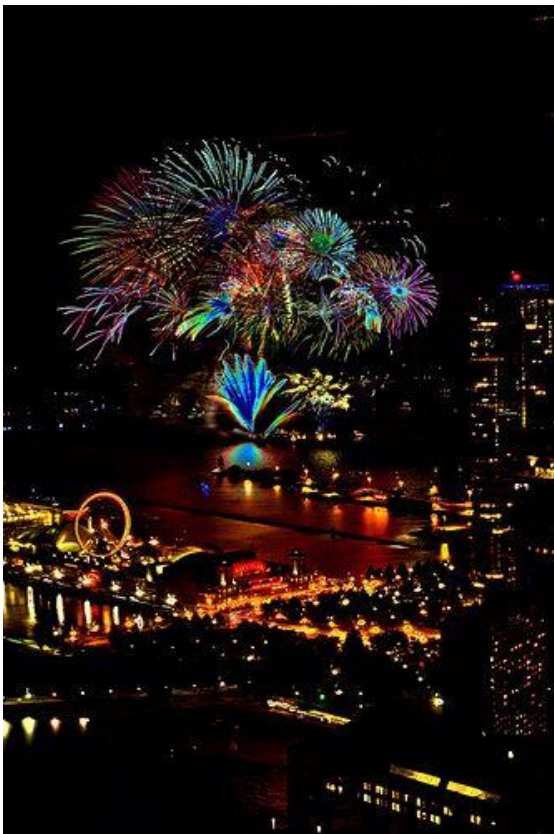
Flow Chart



Sketches



References



<https://www.youtube.com/watch?v=5MDH92VxPEQ>

<https://www.youtube.com/watch?v=23REN8pElcl>